

Self Project

2025-06-03

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# MCP 2.0

A polished MCP 2.0 post that uses normal markdown, guide-style sections, expressive code blocks, GitHub repo cards, tables, and checklists.

## Project Snapshot

### DATE

2025-06-03

### CATEGORY

Self Project

### SHORT DESCRIPTION

A polished MCP 2.0 post that uses normal markdown, guide-style sections, expressive code blocks, GitHub repo cards, tables, and checklists.

### TAGS / TECH STACK

MCP

Protocols

AI Infrastructure

gRPC

Protobuf

Agents

Demo

## Project Links

[GitHub Repository](#)

## Full Project Content

### Project Repository Card

GitHub Repository: [anubhaparashar/MCP2.0](#)

### Introduction

This post is designed to showcase the different kinds of content display supported by this blog theme in one place.

It combines:

- normal markdown writing
- guide-style technical explanation
- expressive code blocks
- a GitHub repository card
- tables and checklists

The project featured here is **MCP 2.0**, a concept that explores what a stronger, more production-ready Model Context Protocol could look like.

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## Why This Project Matters

As AI systems become more complex, protocol design starts to matter more.

MCP 2.0 explores how AI-to-tool communication can become:

- more strongly typed
- more stream-friendly
- easier to discover
- more secure
- more reusable across agents

The main value of this project is not just implementation.  
It is the systems thinking behind the protocol design.

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## Guide-Style Walkthrough

### What problem is it solving?

Traditional tool-calling and context-sharing approaches often become fragile at scale.

Common pain points include:

1. weak typing
2. awkward streaming support
3. no native service discovery
4. broad security scopes
5. custom integration glue for every workflow

## What does MCP 2.0 propose?

MCP 2.0 pushes the protocol toward a more structured model with:

- typed RPC
- capability-based security
- middleware hooks
- discovery services
- multi-agent delegation
- event-driven interaction patterns

## Markdown Elements Showcase

### Checklist

- [x] Add frontmatter
- [x] Add GitHub repo card
- [x] Add expressive code block
- [x] Add architecture diagram
- [x] Add video embed
- [ ] Add custom cover image later

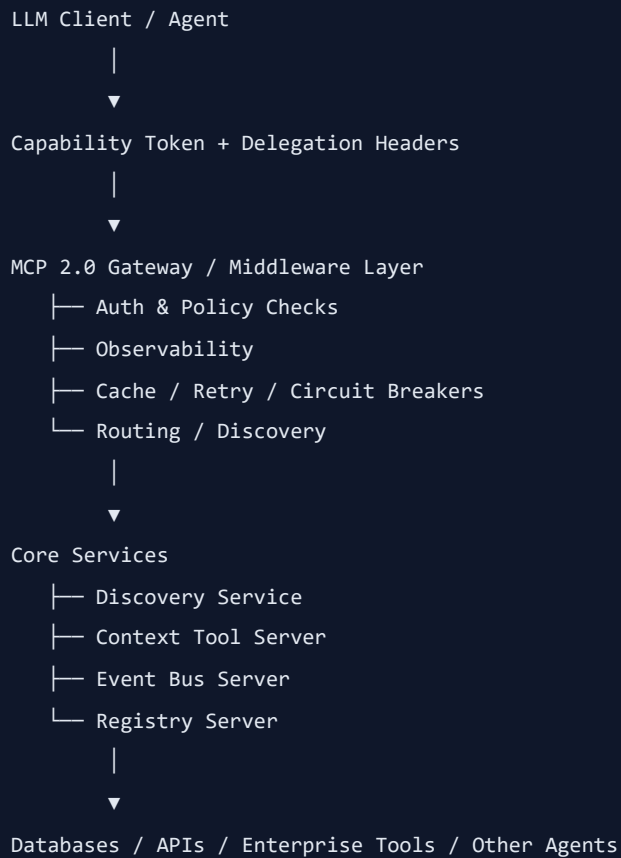
### Table

| Area           | MCP Today    | MCP 2.0 Direction        |
|----------------|--------------|--------------------------|
| Typing         | Looser       | Stronger typed contracts |
| Streaming      | Limited      | First-class              |
| Discovery      | Manual       | Dynamic                  |
| Security       | Broad scopes | Capability-based         |
| Agent chaining | Ad hoc       | Protocol-aware           |

### Quote

Good protocol design reduces integration friction before it becomes a platform problem.

## Proposed Architecture



This view shows that MCP 2.0 is not only a transport idea.  
It is trying to become a protocol layer for governed, scalable AI interactions.

## Repository Structure

```
MCP2.0/
├─ README.md
├─ auth.py
├─ client_example.py
├─ context_tool_server.py
├─ event_bus_server.py
├─ init_db.sql
├─ middleware.py
├─ protos/
│   └─ mcp2.proto
├─ registry_server.py
└─ requirements.txt
```

## File Walkthrough

- [README.md](#) — explains the motivation and protocol vision
- [auth.py](#) — auth and access control concepts
- [middleware.py](#) — reusable cross-cutting protocol behavior

- **registry\_server.py** — service registration and lookup
  - **context\_tool\_server.py** — context and tool interaction layer
  - **event\_bus\_server.py** — event-driven communication support
  - **client\_example.py** — client-side usage example
  - **init\_db.sql** — setup script
  - **requirements.txt** — dependencies
  - **protos/mcp2.proto** — typed protocol contract
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## Expressive Code Block Showcase

The blog theme styles fenced code blocks automatically.

```
syntax = "proto3";

package mcp2;

service Discovery {
  rpc Register(RegisterRequest) returns (RegisterResponse);
  rpc Lookup(LookupRequest) returns (LookupResponse);
}
```

And here is a Python example:

```
class CapabilityToken:
    def __init__(self, subject, allowed_methods):
        self.subject = subject
        self.allowed_methods = allowed_methods

    def can_call(self, method_name: str) -> bool:
        return method_name in self.allowed_methods

token = CapabilityToken("agent-a", ["lookup", "register"])
print(token.can_call("lookup"))
```

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## Design Goals

The main design goals of MCP 2.0 can be summarized as:

- **typed RPC**
- **native streaming**
- **dynamic discovery**
- **fine-grained capability security**

- **pluggable middleware**
- **multi-agent delegation**

This makes the project interesting not only as code, but as a protocol design exercise.

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## References

- Repository: [anubhaparashar/MCP2.0](#)
- Key files: `README.md`, `auth.py`, `client_example.py`, `context_tool_server.py`, `event_bus_server.py`, `init_db.sql`, `middleware.py`, `protos/mcp2.proto`, `registry_server.py`, `requirements.txt`

Source: [src/content/posts/my-first-post.md](#)